KCE PATHWAYS · GRADES 4–5 · UNIT 5: CHARTING THE COURSE · WEEK 24 · MATH

The Bridge Design Studio

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: Shapes in Bridges

Ximena is spending the week at a bridge design studio in Pittsburgh, Pennsylvania. Warm up before the lesson, and show your work. Some problems may feel new, and that is okay!

1. Review (Week 23): a line plot of bolt lengths shows 3 X's at 3/8 inch and 2 X's at 3/4 inch. a) What is the total length of all 5 bolts? b) How much longer is the longest bolt than the shortest?

2. Draw a triangle that has one right angle. Mark the right angle with a small square.

3. How many sides and how many angles does a quadrilateral have? Name two quadrilaterals you already know.

4. Which kind of lines never cross, no matter how long they get?

   parallel lines | perpendicular lines | intersecting lines

5. Draw a shape that has exactly one line of symmetry. Draw the line of symmetry as a dashed line.

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POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Challenge: Checking the Design

Name every triangle two ways, use the family tree, and show your work.

1. Classify each triangle by its sides AND by its angles. a) sides of 6, 6, and 6 feet b) angles of 90°, 30°, and 60°, with no equal sides c) angles of 110°, 35°, and 35°, with two equal sides

2. Two steps: an isosceles triangle brace has a perimeter of 31 feet. Its third side, the one that is not equal to the others, is 7 feet. How long is each equal side?

3. How many lines of symmetry? a) a square b) a rectangle that is not a square c) an equilateral triangle d) a scalene triangle Write your answers, and draw the lines for a) and b).

4. Find the mistake: Xander says every rectangle is a square, because both shapes have four right angles. In two or three complete sentences, explain his mistake.

★ **Challenge:** Write always, sometimes, or never. a) A rectangle is a square. b) A rhombus is a parallelogram. c) A triangle has a pair of parallel sides. Then draw a right trapezoid and mark its parallel sides with arrows.

Looking Ahead: Week 25: Unit 5 Checkpoint. Review the coordinate plane and number patterns, measurement conversions, line plots with fractions, and classifying two-dimensional figures.